

Practica 3: filtro binario

Multimedia

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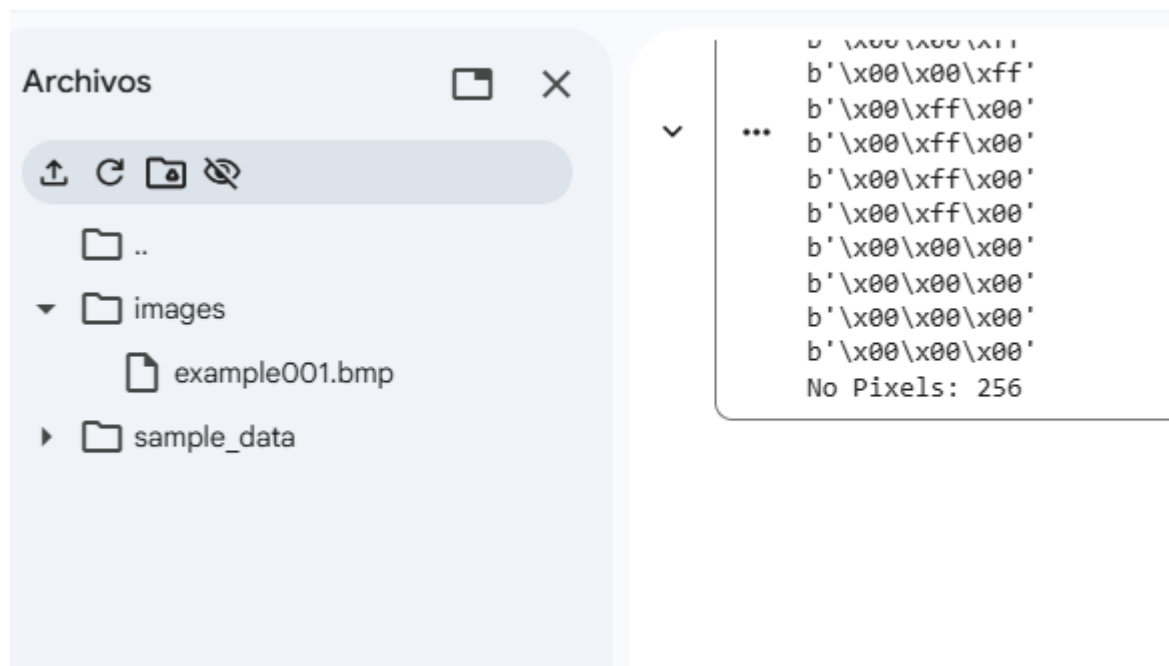
Paso 1 Correr el código:

```
#Read and display data from BMP

file = open("./images/example001.bmp","rb")

firm = file.read(2)
print(firm)
file.seek(54,0)
pixel_first = file.read(3)
print(pixel_first)

file.seek(54,0)
no_pix = 0
while(True):
    pixel_data = file.read(3)
    if(len(pixel_data) > 0):
        print(pixel_data)
        no_pix += 1
    else:
        break
print('No Pixels: '+str(no_pix))
file.close()
```

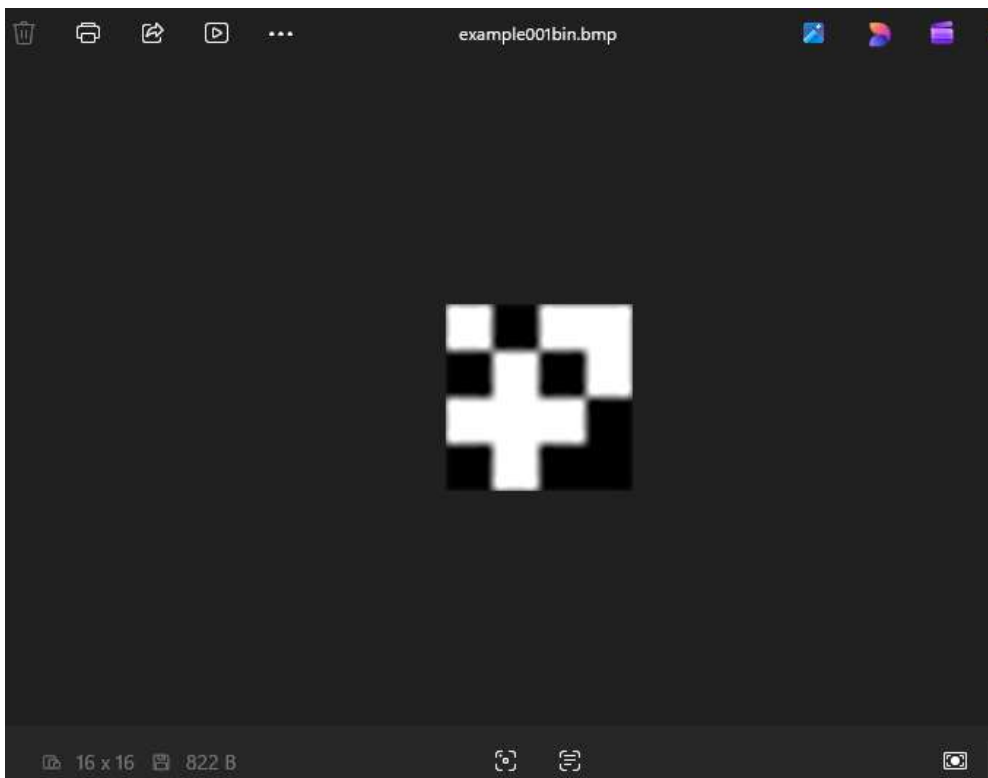


Paso 2 filtro binario

#Binary filter: Black and White

```
file = open('./images/example001.bmp','rb')
fileo = open('./images/example001bin.bmp','wb')
metadata = file.read(54)
fileo.write(metadata)
blanco = [0xff,0xff,0xff]
negro = [0x00,0x00,0x00]

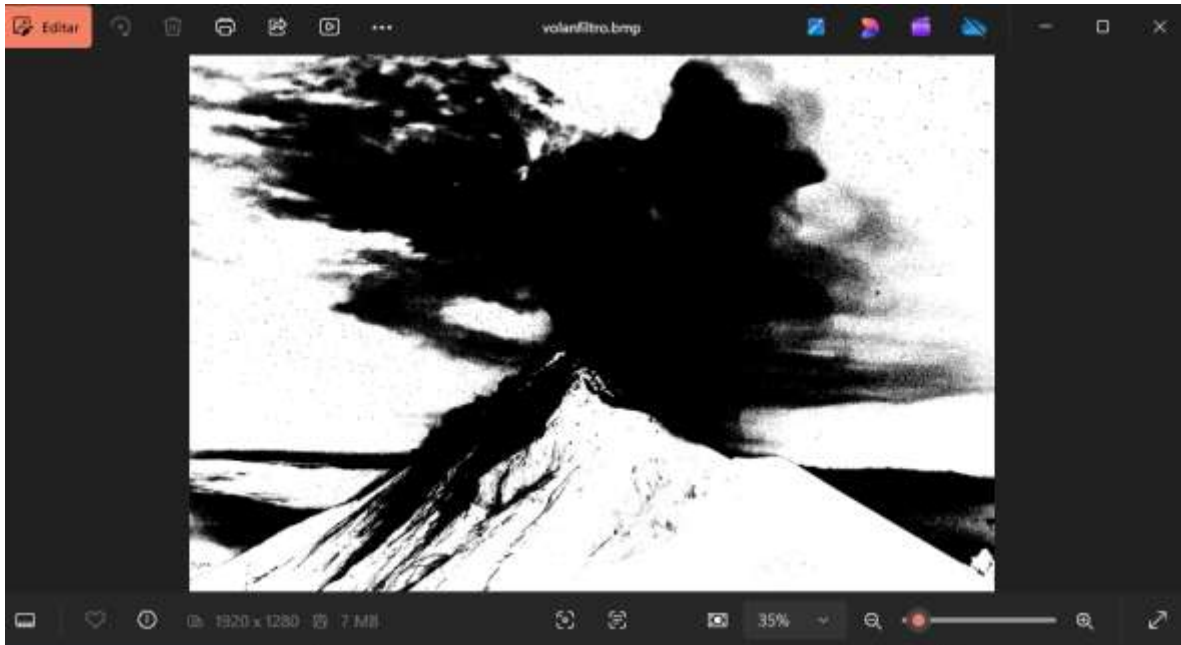
file.seek(54,0)
no_pix = 0
limite = (pow(2, 24)-1)/2
while(True):
    pixel_data = file.read(3)
    if(len(pixel_data) > 0):
        valor_int = int.from_bytes(bytes(pixel_data),byteorder='little')
        if(valor_int<limite):
            fileo.write(bytes(blanco))
        else:
            fileo.write(bytes(negro))
        no_pix += 1
    else:
        break
print('No Pixels: '+str(no_pix))
file.close()
fileo.close()
```



Paso 3

Cambiamos la ruta y generamos el volcán

```
file = open('./images/volcan.bmp', 'rb')  
fileo = open('./images/volanfiltro.bmp', 'wb')  
... ..
```



Paso 4 Invertir el filtro

Cambiamos de orden la asignación de colores

```
negro = [0xff,0xff,0xff]  
blanco = [0x00,0x00,0x00]
```

Y nos da el filtro negro/blanco

